Power Series and Interval of Convergence

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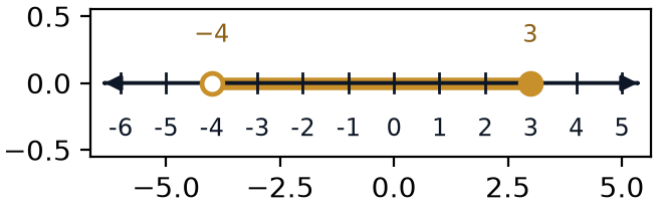
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Finding the radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.

2 [Foundational] • Target: Finding the radius of convergence

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.



3 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.

4 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

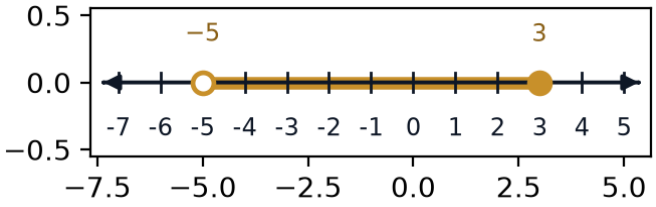
Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$.

5 [Foundational] • Target: Representing functions as power series

Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.

6 [Foundational] • Target: Representing functions as power series

Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.



7 [Foundational] • Target: Differentiating and integrating power series

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Differentiating and integrating power series

Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.

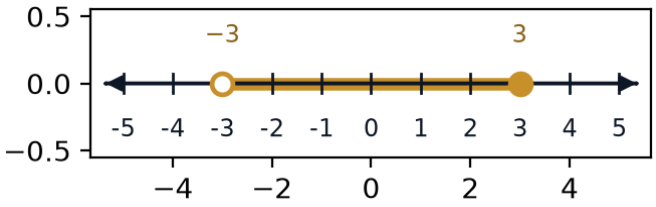
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9 [Intermediate] • Target: More radius/interval of convergence practice

Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n!\, x^n$.

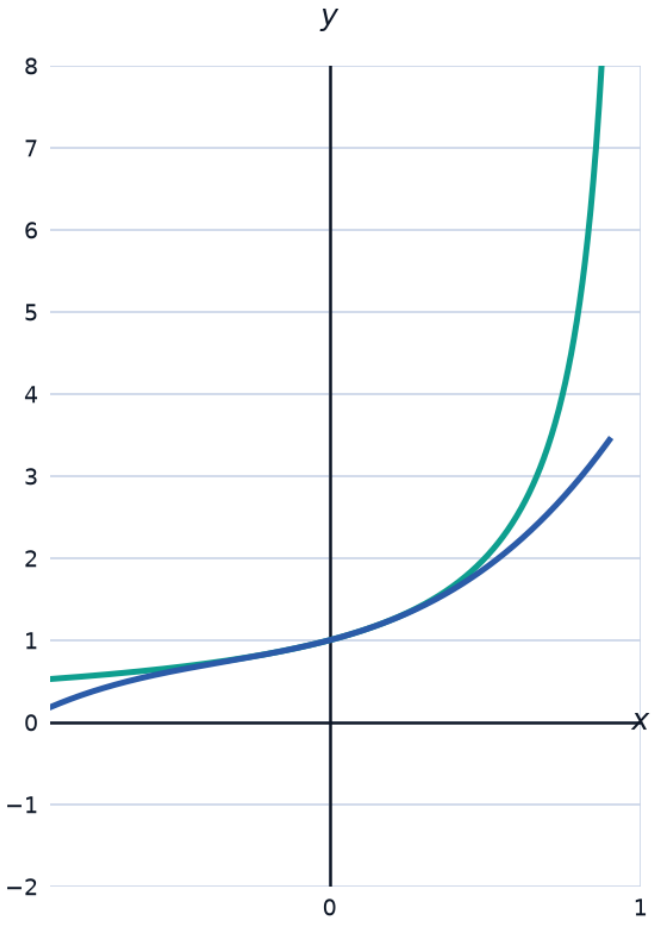
10 [Intermediate] • Target: More radius/interval of convergence practice

Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n \cdot 3^n}$.



11 [Intermediate] • Target: Visualizing a power series approximation

The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.



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12 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.

13 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

14 [Intermediate] • Target: Building a new series by combining known ones

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

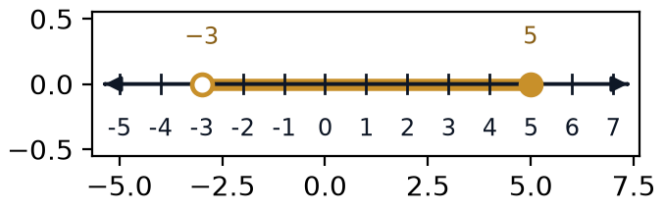
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 3.01 Parametric Equations

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n x^n}{n^2}$.

16 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 3.02 Vector-Valued Functions

For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.



17 [Challenging] • Target: Using a power series to approximate a definite integral • Cross-Lesson: 4.01 Polar Curves and Derivatives

Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

18 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 4.02 Area in Polar Coordinates

Evaluate $\sum_{n=0}^{\infty} 3(\frac{1}{4})^n$.

19 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Evaluate $\int \frac{1}{1+x^2}\,dx$.

20 [Challenging] • Target: Visualizing a power series approximation • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Does $\sum_{n=0}^{\infty} (0.9)^n$ converge, and to what?

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding the radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{6^n}$.

a. 12

b. $\frac{1}{6}$

c. 6

d. 1

22 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{18^n}$.

a. 1

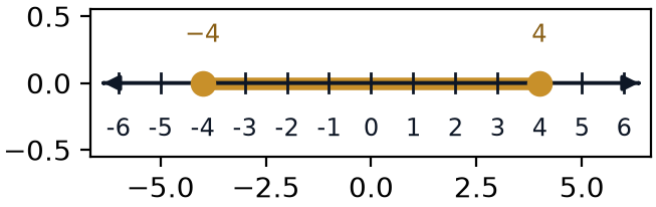
b. 18

c. 36

d. $\frac{1}{18}$

23 [Foundational] • Target: Representing functions as power series

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{5^n}$.



a. 1

b. 10

c. 5

d. $\frac{1}{5}$

24 [Foundational] • Target: Differentiating and integrating power series

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{20^n}$.

a. 1

b. 20

c. 40

d. $\frac{1}{20}$

25 [Foundational] • Target: More radius/interval of convergence practice

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{14^n}$.

a. 28

b. 14

c. $\frac{1}{14}$

d. 1

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26 [Foundational] • Target: Visualizing a power series approximation

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{4^n}$.

a. 4

b. $\frac{1}{4}$

c. 1

d. 8

27 [Foundational] • Target: Term-by-term operations preserve the radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{15^n}$.

a. 1

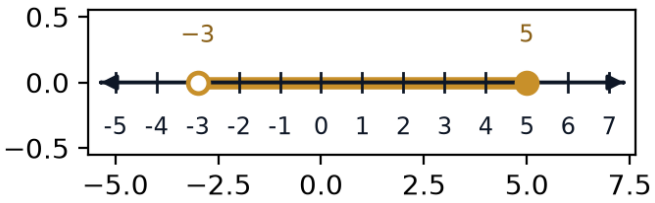
b. $\frac{1}{15}$

c. 15

d. 30

28 [Foundational] • Target: Building a new series by combining known ones

Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+6)^n$.



a. $\frac{1}{7}$

b. 7

c. $\frac{8}{7}$

d. $\frac{1}{8}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Ratio Test with factorial and power combinations

Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+3)^n$.

a. 7

b. $\frac{8}{7}$

c. $\frac{1}{7}$

d. $\frac{1}{8}$

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30 [Intermediate] • Target: Using a power series to approximate a definite integral

Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+7)^n$.

a. $\frac{1}{3}$

b. $\frac{3}{2}$

c. $\frac{1}{2}$

d. 2

31 [Intermediate] • Target: Finding the Radius of Convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+4)^n$.

a. $\frac{1}{2}$

b. 2

c. $\frac{3}{2}$

d. $\frac{1}{3}$

32 [Intermediate] • Target: Testing Interval Endpoints

Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+8)^n$.

a. $\frac{1}{2}$

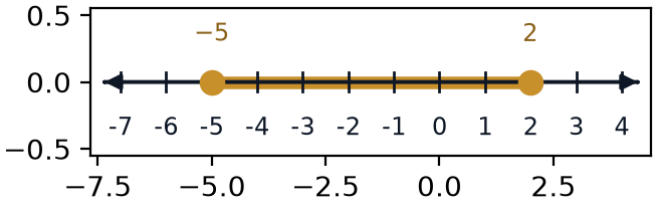
b. $\frac{3}{2}$

c. $\frac{1}{3}$

d. 2

33 [Intermediate] • Target: Term-by-Term Differentiation of Power Series

Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+8)^n$.



a. $\frac{8}{7}$

b. $\frac{1}{7}$

c. 7

d. $\frac{1}{8}$

34 [Intermediate] • Target: Term-by-Term Integration of Power Series

Find the radius of convergence of $\sum_{n=1}^{\infty} 6^n(x+3)^n$.

a. $\frac{1}{6}$

b. 6

c. $\frac{1}{7}$

d. $\frac{7}{6}$

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35 [Intermediate] • Target: Finding the radius of convergence

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-1)^n}{4^n n}$.

a. [-3,5)

b. (-3,5]

c. (-3,5)

d. [-3,5]

36 [Intermediate] • Target: Finding the interval of convergence (testing endpoints)

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-5)^n}{4^n n}$.

a. (1,9]

b. [1,9)

c. [1,9]

d. (1,9)

37 [Intermediate] • Target: Representing functions as power series

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 8^n x^n}{16^n}$.

a. 2

b. 3

c. $\frac{1}{2}$

d. 128

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 4.02 Area in Polar Coordinates

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 6^n x^n}{11^n}$.

a. $\frac{6}{11}$

b. $\frac{17}{6}$

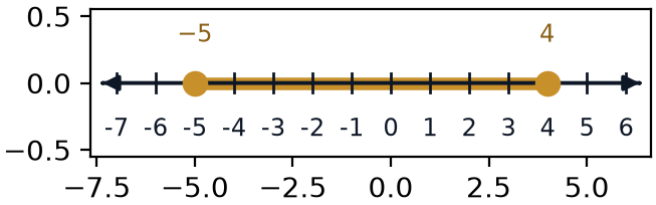
c. $\frac{11}{6}$

d. 66

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39 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^5.



a. 81

b. 729

c. 15

d. 243

40 [Challenging] • Target: Visualizing a power series approximation • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Using $\frac{1}{1-5x} = \sum_{n=0}^{\infty} 5^n x^n$, find the coefficient of x^3.

a. 25

b. 125

c. 15

d. 625

41 [Challenging] • Target: Finding the Radius of Convergence • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the radius of convergence R for the power series $\sum_{n=0}^{\infty} \frac{(x-3)^n}{4^n}$.

a. $R = 4$

b. $R = 1$

c. $R = 3$

d. $R = \infty$

42 [Challenging] • Target: Finding the Radius of Convergence • Cross-Lesson: 1.01 Integration by Parts

Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

a. $R = 0$ (converges only at $x = 0$)

b. $R = 1$

c. $R = \infty$

d. $R = \frac{1}{e}$

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43 [Challenging] • Target: Testing Interval Endpoints • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Why must endpoints always be tested individually when finding the interval of convergence?

a. The Ratio Test yields $L = 1$ at the boundary points $|x - c| = R$ and is inconclusive

b. Endpoints always diverge

c. Endpoints always converge

d. To find the center of the series

44 [Challenging] • Target: Term-by-Term Differentiation of Power Series • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Starting from $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series representation for $\frac{1}{(1-x)^2}$.

a. $\sum_{n=1}^{\infty} nx^{n-1}$

b. $\sum_{n=0}^{\infty} x^{2n}$

c. $\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$

d. $\sum_{n=1}^{\infty} n^2 x^n$

45 [Challenging] • Target: Term-by-Term Integration of Power Series • Cross-Lesson: 3.01 Parametric Equations

Integrating $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$ term-by-term yields the Maclaurin series for which function?

a. $\ln(1+x)$

b. $\arctan x$

c. e^{-x}

d. $\frac{1}{(1+x)^2}$

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